SEQUENCES

DEF:

A **sequence** is a list of numbers in a given order:

$$\{a_1,\ a_2,\ \cdots,\ a_n,\ \cdots\cdots\}$$

Example $1,\ \frac{1}{2},\ \frac{1}{3},\ \frac{1}{4},\ \frac{1}{5},\cdots\cdots\cdots$ $\left\{\frac{1}{n}\right\}_{n=1}^{\infty}$ $a_n = \frac{1}{n}$

first term

second term

n-th term

index

Example $\frac{1}{2},\ \frac{2}{3},\ \frac{3}{4},\ \frac{4}{5},\cdots\cdots\cdots$ $\left\{\frac{n}{n+1}\right\}_{n=1}^{\infty}$

Example $\frac{2}{3},\ \frac{3}{9},\ \frac{4}{27},\ \frac{5}{81},\cdots\cdots\cdots$ $\left\{\frac{n}{3^{n-1}}\right\}_{n=2}^{\infty}$

SEQUENCES

Example $\left\{-\frac{1}{2},\ \frac{2}{3},\ -\frac{3}{4},\ \frac{4}{5},\ -\frac{5}{6},\ldots\ldots\right\}$ $\left\{(-1)^n\frac{n}{n+1}\right\}_{n=1}^{\infty}$

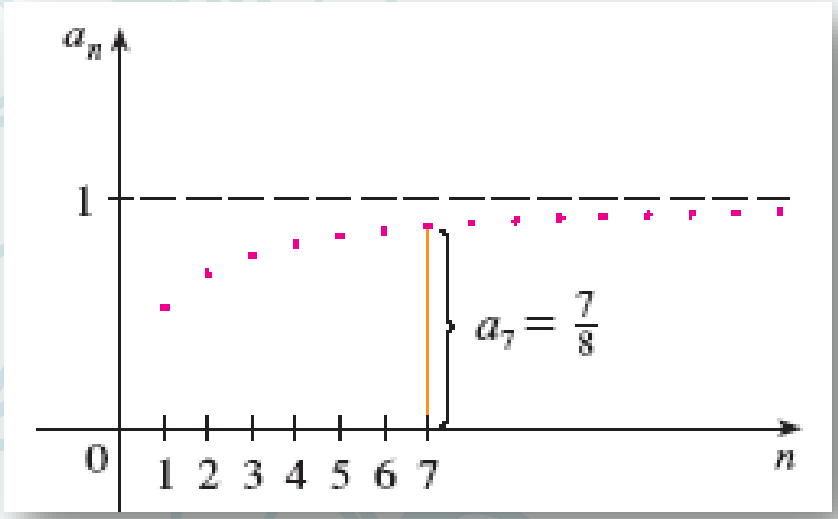
Example $\left\{-\frac{2}{3},\ \frac{3}{9},\ -\frac{4}{27},\ \frac{5}{81},\ldots\ldots\right\}$ $\left\{(-1)^{n-1}\frac{n}{3^{n-1}}\right\}_{n=2}^{\infty}$

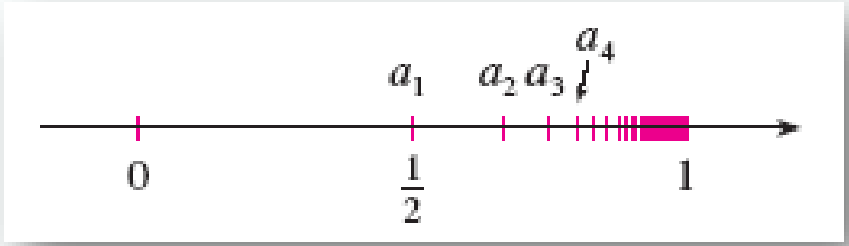
Example Find a formula for the general term of the sequence

$$\left\{\frac{11}{5}, -\frac{13}{25}, \frac{15}{125}, -\frac{17}{625}, \ldots\ldots\right\}$$

PLOT THE SEQUENCES

Example $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \ldots\ldots\ldots$ $\left\{\frac{n}{n+1}\right\}_{n=1}^{\infty}$

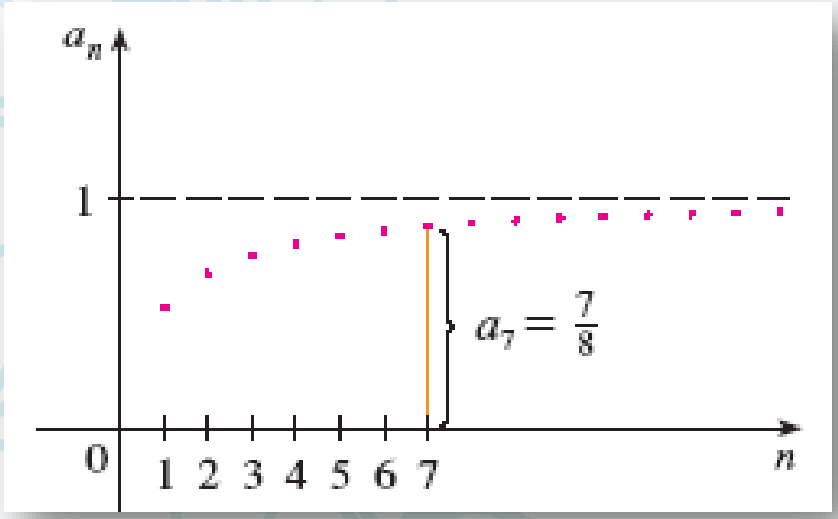


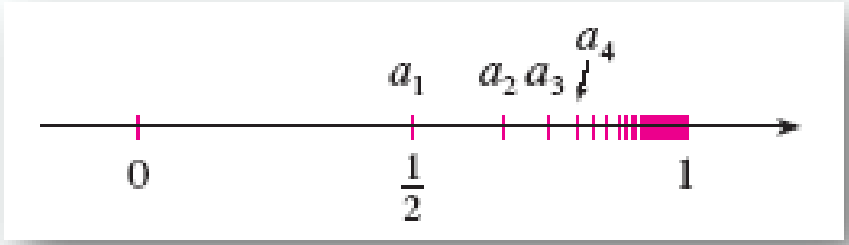


LIMIT OF THE SEQUENCES

Example

$$\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \ldots\ldots\ldots \qquad \left\{\frac{n}{n+1}\right\}_{n=1}^{\infty}$$





Example

$$\lim_{n\to\infty} a_n = 1 \qquad \lim_{n\to\infty} \frac{n}{n+1} = 1$$

SEQUENCES

1 Definition A sequence $\{a_n\}$ has the **limit** L and we write

$$\lim_{n \to \infty} a_n = L \qquad \text{or} \qquad a_n \to L \text{ as } n \to \infty$$

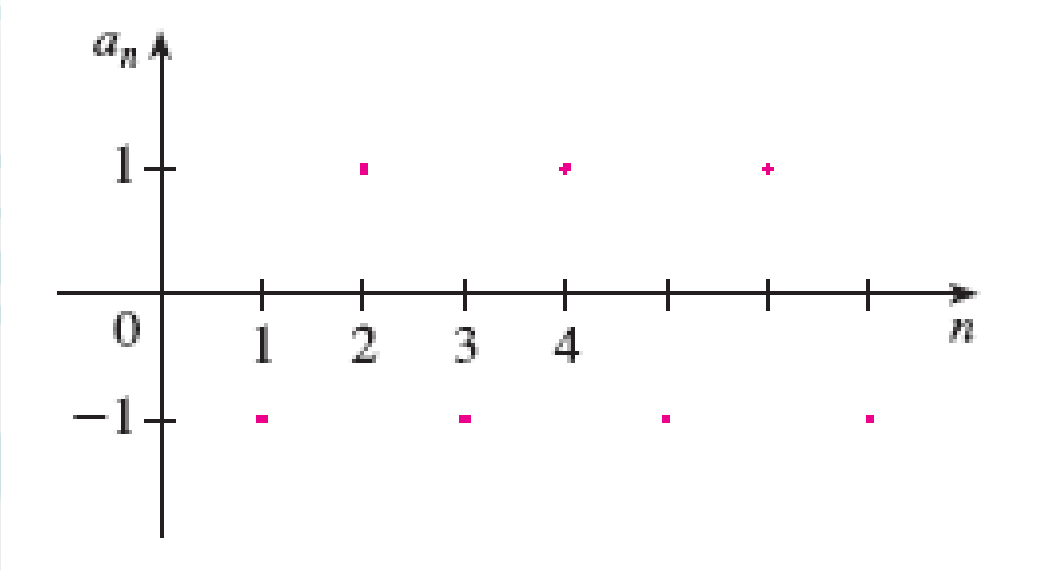
if we can make the terms a_n as close to L as we like by taking n sufficiently large. If $\lim_{n \to \infty} a_n$ exists, we say the sequence **converges** (or is **convergent**). Otherwise, we say the sequence **diverges** (or is **divergent**).

THEOREM 1 Let $\{a_n\}$ and $\{b_n\}$ be sequences of real numbers, and let A and B be real numbers. The following rules hold if $\lim_{n\to\infty} a_n = A$ and $\lim_{n\to\infty} b_n = B$.

1. *Sum Rule:* $\lim_{n\to\infty}(a_n + b_n) = A + B$
2. *Difference Rule:* $\lim_{n\to\infty}(a_n - b_n) = A - B$
3. *Constant Multiple Rule:* $\lim_{n\to\infty}(k \cdot b_n) = k \cdot B$ (any number k)
4. *Product Rule:* $\lim_{n\to\infty}(a_n \cdot b_n) = A \cdot B$
5. *Quotient Rule:* $\lim_{n\to\infty} \dfrac{a_n}{b_n} = \dfrac{A}{B}$ if $B \neq 0$

Example Determine whether the sequence is convergent or divergent.

$$\lim_{n\to\infty}\left\{(-1)^n\right\}$$



6 THEOREM If $\lim_{n\to\infty} |a_n| = 0$, then $\lim_{n\to\infty} a_n = 0$.

Example

$$\lim_{n\to\infty}\left((-1)^n \frac{1}{n}\right)$$

Note:

$$\lim_{n\to\infty}\left((-1)^n \frac{n}{n+1}\right)$$

EXAMPLE 5 Is the sequence $a_n = \dfrac{n}{\sqrt{10 + n}}$ convergent or divergent?

Example: Find $\lim_{n \to \infty} \left(\dfrac{n}{n+1} \right)$

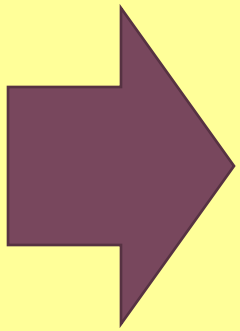
Example: Find $\lim_{n\to\infty} \ln n/n$

THEOREM;

$f(x)$ **continuous**

$\{a_n\}$ **convergent**

$a_n \to L$



$\{f(a_n)\}$ **convergent**

$f(a_n) \to f(L)$

Example Find

$$\lim_{n\to\infty} \sin\left(\frac{\pi}{n}\right)$$

SEQUENCES

Factorial;

$$n! = 1 \cdot 2 \cdot 3 \cdots (n-1) \cdot n$$

NOTE

$$3! = 3 \times 2 \times 1 = 6$$
$$10! = 10 \times (9!)$$
$$n! = n \times (n-1)!$$

Example

Find $\lim_{n \to \infty} (n+1)!/n!$

THEOREM; **(SQUEEZE THEOREM FOR SEQUENCES)**

$$a_n \le b_n \le c_n \quad \text{for} \quad n \ge n_0$$

$$a_n \to L \qquad c_n \to L$$

$$\Rightarrow \quad b_n \to L$$

Example:Find

$$\lim_{n\to\infty}\left\{\frac{\cos n}{n}\right\}$$

$$\lim_{n\to\infty}\{5^n\}$$

$$\lim_{n\to\infty}\left\{\frac{1}{2^n}\right\}$$

Example For what values of r is the sequence convergent?

$$\{ r^n \} \qquad \lim_{n \to \infty} r^n$$

The sequence $\{ r^n \}$ is $\begin{cases} \text{conv} & -1 < r \leq 1 \\ \text{div} & \text{other values} \end{cases}$

$$\lim_{n \to \infty} r^n = \begin{cases} 0 & \text{if } -1 < r < 1 \\ 1 & \text{if } r = 1 \end{cases}$$

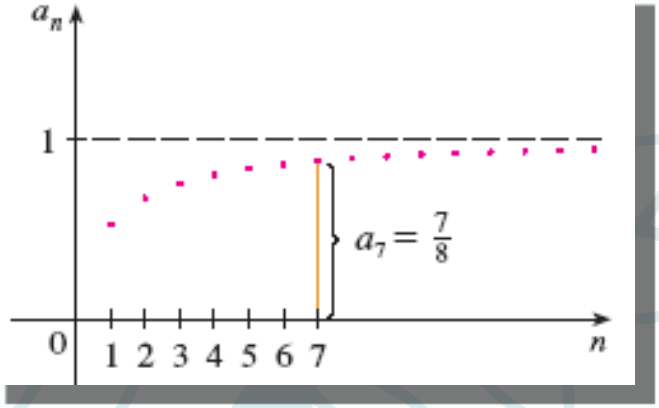
SEQUENCES

DEFINITION

$\{a_n\}$ **bounded from above**

$a_n \le M$ for all n

M **Upper bound**

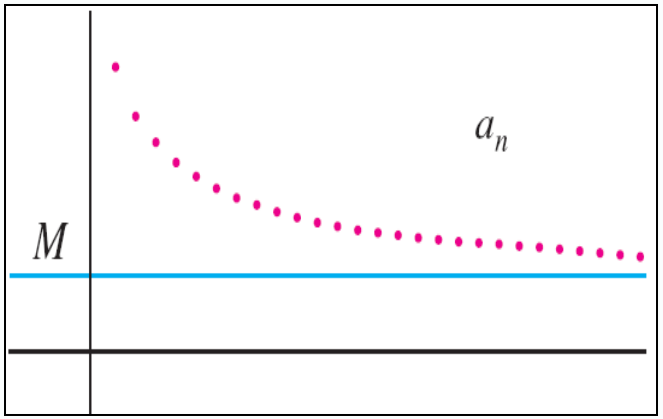


DEFINITION

$\{a_n\}$ **bounded from below**

$a_n \ge M$ for all n

M **Lower bound**



Example

$\left\{\dfrac{n}{n+1}\right\}$ **Is bounded above by any number greater than one**

$M = 1$ **Least upper bound**

Example $\left\{3+\dfrac{1}{n}\right\}$ **Is bounded below**

$a_n > 3$

If $\{a_n\}$ is bounded from above and below,

$\Rightarrow$

$\{a_n\}$ **bounded**

If $\{a_n\}$ is not bounded

$\Rightarrow$ we say that

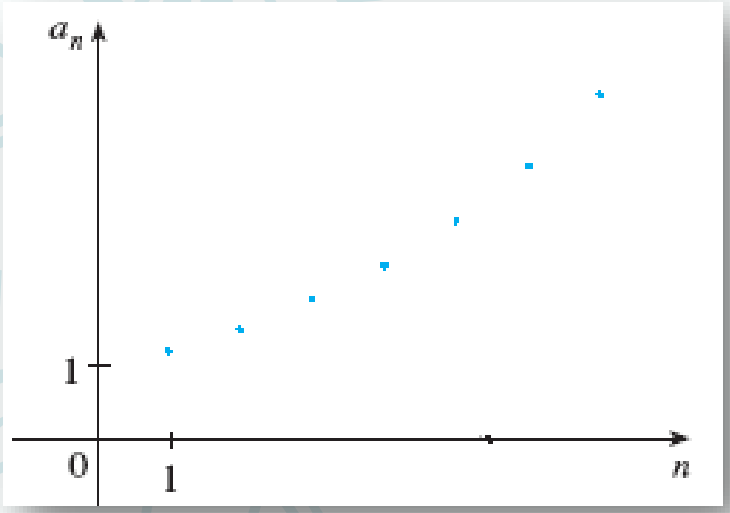
$\{a_n\}$ **unbounded**

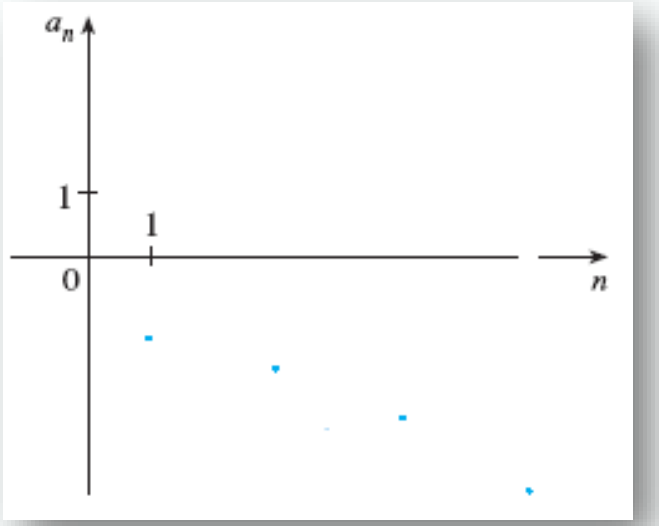
SEQUENCES

DEFINITION

$\{a_n\}$ **Increasing if** $a_n \le a_{n+1}$ for all $n \ge 1$

$$a_1 \le a_2 \le a_3 \le a_4 \le \cdots\cdots$$





DEFINITION

$\{a_n\}$ **Is decreasing if** $a_n \ge a_{n+1}$ for all $n \ge 1$

$$a_1 \ge a_2 \ge a_3 \ge a_4 \ge \cdots\cdots$$

DEFINITION $\{ a_n \}$ **monotonic**

if it is either nonincreasing or nondecreasing.

SEQUENCES

Example Is the sequence increasing or decreasing $\left\{ \frac{3}{n+5} \right\}$

Example12 Is the sequence increasing or decreasing $\left\{ \frac{n}{n^2+1} \right\}$

THEOREM 6—The Monotonic Sequence Theorem If a sequence $\{a_n\}$ is both bounded and monotonic, then the sequence converges.

How to find a limit of a sequence

3 THEOREM If $\lim_{x\to\infty} f(x) = L$ and $f(n) = a_n$ when n is an integer, then $\lim_{n\to\infty} a_n = L$.

6 THEOREM If $\lim_{n\to\infty} |a_n| = 0$, then $\lim_{n\to\infty} a_n = 0$.

THEOREM; $f(x)$ **continuous** $\{a_n\}$ **convergent** $a_n \to L$ ⇒ $\{f(a_n)\}$ **convergent** $f(a_n) \to f(L)$

THEOREM; (SQUEEZE THEOREM)

$a_n \le b_n \le c_n$ for $n \ge n_0$

$a_n \to L$ $c_n \to L$ ⇒ $b_n \to L$

THEOREM;

Every bounded, monotonic sequence is convergent

The sequence $\{r^n\}$ is $\begin{cases} \text{conv} & -1 < r \le 1 \\ \text{div} & \text{other values} \end{cases}$

$$\lim_{n\to\infty} r^n = \begin{cases} 0 & \text{if } -1 < r < 1 \\ 1 & \text{if } r = 1 \end{cases}$$

SEQUENCES

Math 102 (Term 101) Final Exam

The sequence $\left\{\frac{\ln(n^3)}{n}\right\}_{n=1}^{\infty}$

(a) is convergent and its limit is 0

(b) is convergent and its limit is 1

(c) is convergent and its limit is 3

(d) is convergent and its limit is $\frac{-1}{3}$

(e) is divergent

Math 102, Final Exam, Term 091

The sequence $\left\{ \dfrac{(2n-1)! \cdot (5n^2+2n)}{(2n+1)!} \right\}_{n=1}^{+\infty}$

(a) converges and its limit is $\dfrac{5}{2}$

(b) converges and its limit is 5

(c) converges and its limit is 1

(d) converges and its limit is $\dfrac{5}{4}$

(e) is divergent

SEQUENCES

Math 102 Final Exam Term 112

The sequence $\left\{\frac{4}{5}, \frac{6}{8}, \frac{8}{11}, \frac{10}{14}, \ldots\right\}$

(a) converges to $\frac{1}{3}$

(b) converges to $\frac{2}{5}$

(c) converges to 0

(d) converges to $\frac{2}{3}$

(e) is divergent

Math 102, Final Exam, Term 102

The sequence $\left\{\frac{\ln n}{n^2}\right\}_{n\geq 3}$ is

(a) Increasing and divergent

(b) Decreasing and divergent

(c) Increasing and convergent

(d) Decreasing and convergent

(e) Neither increassing, nor decreasing and convergent

SEQUENCES

Math 102, Final Exam, Term 111

The sequence $\left\{\sqrt[n]{2^{3n-1}}\right\}_{n=2}^{\infty}$ is

(a) convergent and its limits is 4

(b) divergent

(c) convergent and its limits is $1/2$

(d) convergent and its limits is 8

(e) convergent and its limits is 2

Math 102, Final Exam (092)

The limit of the sequence defined by $s_n = \frac{1}{n}\sin\left(\frac{n\pi}{4}\right) + \frac{(2n+6)}{(n+1)}$

(a) is equal to 2

(b) is equal to 0

(c) is equal to 1

(d) oscillates between -1 and 1

(e) is ∞

SEQUENCES

17–46 Determine whether the sequence converges or diverges. If it converges, find the limit.

17. $a_n = 1 - (0.2)^n$

18. $a_n = \dfrac{n^3}{n^3 + 1}$

19. $a_n = \dfrac{3 + 5n^2}{n + n^2}$

20. $a_n = \dfrac{n^3}{n + 1}$

21. $a_n = e^{1/n}$

22. $a_n = \dfrac{3^{n+2}}{5^n}$

23. $a_n = \tan\left(\dfrac{2n\pi}{1 + 8n}\right)$

24. $a_n = \sqrt{\dfrac{n + 1}{9n + 1}}$

25. $a_n = \dfrac{(-1)^{n-1}n}{n^2 + 1}$

26. $a_n = \dfrac{(-1)^n n^3}{n^3 + 2n^2 + 1}$

27. $a_n = \cos(n/2)$

28. $a_n = \cos(2/n)$

29. $\left\{\dfrac{(2n - 1)!}{(2n + 1)!}\right\}$

30. $\{\arctan 2n\}$

31. $\left\{\dfrac{e^n + e^{-n}}{e^{2n} - 1}\right\}$

32. $\left\{\dfrac{\ln n}{\ln 2n}\right\}$

33. $\{n^2 e^{-n}\}$

34. $\{n \cos n\pi\}$

35. $a_n = \dfrac{\cos^2 n}{2^n}$

36. $a_n = \ln(n + 1) - \ln n$

37. $a_n = n \sin(1/n)$

38. $a_n = \sqrt[n]{2^{1+3n}}$

39. $a_n = \left(1 + \dfrac{2}{n}\right)^n$

40. $a_n = \dfrac{\sin 2n}{1 + \sqrt{n}}$

41. $a_n = \ln(2n^2 + 1) - \ln(n^2 + 1)$

43. $\{0, 1, 0, 0, 1, 0, 0, 0, 1, \ldots\}$

45. $a_n = \dfrac{n!}{2^n}$

42. $a_n = \dfrac{(\ln n)^2}{n}$

44. $\left\{\frac{1}{1}, \frac{1}{3}, \frac{1}{2}, \frac{1}{4}, \frac{1}{3}, \frac{1}{5}, \frac{1}{4}, \frac{1}{6}, \ldots\right\}$

46. $a_n = \dfrac{(-3)^n}{n!}$

SEQUENCES

Q.6. Determine whether the sequence converges or diverges. If it converges, find its limit.

(a) . (4 points) . $a_n = \dfrac{1+\sin 2n}{1+\sqrt{n}}$.

Solution. By the squeeze Theorem, and since $0 \leq 1+\sin 2n \leq 2$.

Then, $0 \leq \dfrac{1+\sin 2n}{1+\sqrt{n}} \leq \dfrac{2}{1+\sqrt{n}}$..... (3 points)

So, $\lim\limits_{n\longrightarrow\infty} 0 \leq \lim\limits_{n\longrightarrow\infty} \dfrac{1+\sin 2n}{1+\sqrt{n}} \leq \lim\limits_{n\longrightarrow\infty} \dfrac{2}{1+\sqrt{n}}$.

Thus, $0 \leq \lim\limits_{n\longrightarrow\infty} \dfrac{1+\sin 2n}{1+\sqrt{n}} \leq 0$.

$\Longrightarrow \lim\limits_{n\longrightarrow\infty} \dfrac{1+\sin 2n}{1+\sqrt{n}} = 0$..... (1 point)

CONVERGES.

TERM-082

Q.6. (b) . (4 points) . $a_n = \ln\sqrt{n+1} - \frac{1}{2}\ln n.$

Solution. $\lim_{n\longrightarrow\infty}\left(\ln\sqrt{n+1} - \frac{1}{2}\ln n\right)$

$= \lim_{n\longrightarrow\infty}\left(\frac{1}{2}\left[\ln(n+1) - \ln n\right]\right)$

$= \lim_{n\longrightarrow\infty}\frac{1}{2}\ln\frac{(n+1)}{n}$..... (2 points)

$= \frac{1}{2}\ln\left[\lim_{n\longrightarrow\infty}\left(\frac{n+1}{n}\right)\right]$

$= \frac{1}{2}\ln 1$

$= 0$..... (2 points)

CONVERGES.

TERM-082

Q.9. $(6 - \ points)$. Determine whether the sequence

$$\left(\frac{2}{4}-\frac{1}{3}\right), \left(\frac{4}{5}-\frac{1}{5}\right), \left(\frac{6}{6}-\frac{1}{7}\right), \left(\frac{8}{7}-\frac{1}{9}\right), \ldots$$

converges or diverges. If it converges, find the limit.

TERM-092

SOLUTION. $a_n = \dfrac{2n}{n+3} - \dfrac{1}{2n+1}$ ..... $\longrightarrow (3 - points)$

$$= \frac{2}{1+3/n} - \frac{1}{2n+1}$$

$$\lim_{n \longrightarrow \infty} a_n = \frac{2}{1+0} - 0 = 2 \text{ ..... } \longrightarrow (2 - points)$$

The given sequence CONVERGES. ..... $\longrightarrow (1 - point)$

SEQUENCES

Q.3. $(6 - \ points)$. Determine whether the sequence $\left\{\dfrac{e^n + e^{-n}}{e^{2n} - 1}\right\}$ converges or diverges.

If it converges, find the limit.

TERM-092

SOLUTION. Let $a_n = \dfrac{e^n + e^{-n}}{e^{2n} - 1} = \dfrac{e^{-n}\left(e^{2n} + 1\right)}{e^{2n} - 1}$

$= e^{-n}\left(\dfrac{e^{2n} + 1}{e^{2n} - 1}\right) = e^{-n}\left(\dfrac{1 + \dfrac{1}{e^{2n}}}{1 - \dfrac{1}{e^{2n}}}\right)$ .... $\longrightarrow (3 - points)$

$\lim\limits_{n \longrightarrow \infty} a_n = 0\left(\dfrac{1+0}{1-0}\right) = 0$..... $\longrightarrow (2 - points)$

The sequence $\left\{a_n = \dfrac{e^n + e^{-n}}{e^{2n} + 1}\right\}$ CONVERGES. .... $\longrightarrow (1 - point)$

Exercises 10.1

Finding Terms of a Sequence

Each of Exercises 1–6 gives a formula for the nth term a_n of a sequence $\{a_n\}$. Find the values of a_1, a_2, a_3, and a_4.

1. $a_n = \dfrac{1 - n}{n^2}$

2. $a_n = \dfrac{1}{n!}$

3. $a_n = \dfrac{(-1)^{n+1}}{2n - 1}$

4. $a_n = 2 + (-1)^n$

5. $a_n = \dfrac{2^n}{2^{n+1}}$

6. $a_n = \dfrac{2^n - 1}{2^n}$

Each of Exercises 7–12 gives the first term or two of a sequence along with a recursion formula for the remaining terms. Write out the first ten terms of the sequence.

7. $a_1 = 1, \quad a_{n+1} = a_n + (1/2^n)$

8. $a_1 = 1, \quad a_{n+1} = a_n/(n + 1)$

9. $a_1 = 2, \quad a_{n+1} = (-1)^{n+1}a_n/2$

10. $a_1 = -2, \quad a_{n+1} = na_n/(n + 1)$

11. $a_1 = a_2 = 1, \quad a_{n+2} = a_{n+1} + a_n$

12. $a_1 = 2, \quad a_2 = -1, \quad a_{n+2} = a_{n+1}/a_n$

Finding a Sequence's Formula

In Exercises 13–26, find a formula for the nth term of the sequence.

13. The sequence $1, -1, 1, -1, 1, \ldots$ — 1's with alternating signs

14. The sequence $-1, 1, -1, 1, -1, \ldots$ — 1's with alternating signs

15. The sequence $1, -4, 9, -16, 25, \ldots$ — Squares of the positive integers, with alternating signs

16. The sequence $1, -\frac{1}{4}, \frac{1}{9}, -\frac{1}{16}, \frac{1}{25}, \cdots$ — Reciprocals of squares of the positive integers, with alternating signs

17. $\frac{1}{9}, \frac{2}{12}, \frac{2^2}{15}, \frac{2^3}{18}, \frac{2^4}{21}, \cdots$ — Powers of 2 divided by multiples of 3

18. $-\frac{3}{2}, -\frac{1}{6}, \frac{1}{12}, \frac{3}{20}, \frac{5}{30}, \cdots$ — Integers differing by 2 divided by products of consecutive integers

19. The sequence $0, 3, 8, 15, 24, \ldots$ — Squares of the positive integers diminished by 1

20. The sequence $-3, -2, -1, 0, 1, \ldots$ — Integers, beginning with -3

21. The sequence $1, 5, 9, 13, 17, \ldots$ — Every other odd positive integer

22. The sequence $2, 6, 10, 14, 18, \ldots$ — Every other even positive integer

23. $\frac{5}{1}, \frac{8}{2}, \frac{11}{6}, \frac{14}{24}, \frac{17}{120}, \cdots$ — Integers differing by 3 divided by factorials

24. $\frac{1}{25}, \frac{8}{125}, \frac{27}{625}, \frac{64}{3125}, \frac{125}{15{,}625}, \ldots$ Cubes of positive integers divided by powers of 5

25. The sequence 1, 0, 1, 0, 1, . . . Alternating 1's and 0's

26. The sequence 0, 1, 1, 2, 2, 3, 3, 4, . . . Each positive integer repeated

Convergence and Divergence

Which of the sequences $\{a_n\}$ in Exercises 27–90 converge, and which diverge? Find the limit of each convergent sequence.

27. $a_n = 2 + (0.1)^n$

28. $a_n = \dfrac{n + (-1)^n}{n}$

29. $a_n = \dfrac{1 - 2n}{1 + 2n}$

30. $a_n = \dfrac{2n + 1}{1 - 3\sqrt{n}}$

31. $a_n = \dfrac{1 - 5n^4}{n^4 + 8n^3}$

32. $a_n = \dfrac{n + 3}{n^2 + 5n + 6}$

33. $a_n = \dfrac{n^2 - 2n + 1}{n - 1}$

34. $a_n = \dfrac{1 - n^3}{70 - 4n^2}$

35. $a_n = 1 + (-1)^n$

36. $a_n = (-1)^n \left(1 - \dfrac{1}{n}\right)$

37. $a_n = \left(\dfrac{n + 1}{2n}\right)\left(1 - \dfrac{1}{n}\right)$

38. $a_n = \left(2 - \dfrac{1}{2^n}\right)\left(3 + \dfrac{1}{2^n}\right)$

39. $a_n = \dfrac{(-1)^{n+1}}{2n - 1}$

40. $a_n = \left(-\dfrac{1}{2}\right)^n$

41. $a_n = \sqrt{\dfrac{2n}{n + 1}}$

42. $a_n = \dfrac{1}{(0.9)^n}$

43. $a_n = \sin\left(\dfrac{\pi}{2} + \dfrac{1}{n}\right)$

44. $a_n = n\pi \cos(n\pi)$

45. $a_n = \dfrac{\sin n}{n}$

46. $a_n = \dfrac{\sin^2 n}{2^n}$

47. $a_n = \dfrac{n}{2^n}$

48. $a_n = \dfrac{3^n}{n^3}$

49. $a_n = \dfrac{\ln(n + 1)}{\sqrt{n}}$

50. $a_n = \dfrac{\ln n}{\ln 2n}$

51. $a_n = 8^{1/n}$

52. $a_n = (0.03)^{1/n}$

53. $a_n = \left(1 + \frac{7}{n}\right)^n$

54. $a_n = \left(1 - \frac{1}{n}\right)^n$

55. $a_n = \sqrt[n]{10n}$

56. $a_n = \sqrt[n]{n^2}$

57. $a_n = \left(\frac{3}{n}\right)^{1/n}$

58. $a_n = (n + 4)^{1/(n+4)}$

59. $a_n = \frac{\ln n}{n^{1/n}}$

60. $a_n = \ln n - \ln (n + 1)$

61. $a_n = \sqrt[n]{4^n n}$

62. $a_n = \sqrt[n]{3^{2n+1}}$

63. $a_n = \dfrac{n!}{n^n}$ (*Hint:* Compare with $1/n$.)

64. $a_n = \dfrac{(-4)^n}{n!}$

65. $a_n = \dfrac{n!}{10^{6n}}$

66. $a_n = \dfrac{n!}{2^n \cdot 3^n}$

67. $a_n = \left(\dfrac{1}{n}\right)^{1/(\ln n)}$

68. $a_n = \ln\left(1 + \dfrac{1}{n}\right)^n$

69. $a_n = \left(\dfrac{3n+1}{3n-1}\right)^n$

70. $a_n = \left(\dfrac{n}{n+1}\right)^n$

71. $a_n = \left(\dfrac{x^n}{2n+1}\right)^{1/n}, \quad x > 0$

72. $a_n = \left(1 - \dfrac{1}{n^2}\right)^n$

73. $a_n = \dfrac{3^n \cdot 6^n}{2^{-n} \cdot n!}$

74. $a_n = \dfrac{(10/11)^n}{(9/10)^n + (11/12)^n}$

75. $a_n = \tanh n$

76. $a_n = \sinh(\ln n)$

77. $a_n = \dfrac{n^2}{2n-1} \sin\dfrac{1}{n}$

78. $a_n = n\left(1 - \cos\dfrac{1}{n}\right)$

79. $a_n = \sqrt{n} \sin\dfrac{1}{\sqrt{n}}$

80. $a_n = (3^n + 5^n)^{1/n}$

81. $a_n = \tan^{-1} n$

82. $a_n = \dfrac{1}{\sqrt{n}} \tan^{-1} n$

83. $a_n = \left(\dfrac{1}{3}\right)^n + \dfrac{1}{\sqrt{2^n}}$

84. $a_n = \sqrt[n]{n^2 + n}$

85. $a_n = \dfrac{(\ln n)^{200}}{n}$

86. $a_n = \dfrac{(\ln n)^5}{\sqrt{n}}$

87. $a_n = n - \sqrt{n^2 - n}$

88. $a_n = \dfrac{1}{\sqrt{n^2 - 1} - \sqrt{n^2 + n}}$

89. $a_n = \dfrac{1}{n} \displaystyle\int_1^n \frac{1}{x}\,dx$

90. $a_n = \displaystyle\int_1^n \frac{1}{x^p}\,dx, \quad p > 1$

Recursively Defined Sequences

In Exercises 91–98, assume that each sequence converges and find its limit.

91. $a_1 = 2, \quad a_{n+1} = \dfrac{72}{1 + a_n}$

92. $a_1 = -1, \quad a_{n+1} = \dfrac{a_n + 6}{a_n + 2}$

93. $a_1 = -4, \quad a_{n+1} = \sqrt{8 + 2a_n}$

94. $a_1 = 0, \quad a_{n+1} = \sqrt{8 + 2a_n}$

95. $a_1 = 5, \quad a_{n+1} = \sqrt{5a_n}$

96. $a_1 = 3, \quad a_{n+1} = 12 - \sqrt{a_n}$

97. $2, 2 + \dfrac{1}{2}, 2 + \dfrac{1}{2 + \dfrac{1}{2}}, 2 + \dfrac{1}{2 + \dfrac{1}{2 + \dfrac{1}{2}}}, \ldots$

98. $\sqrt{1}, \sqrt{1 + \sqrt{1}}, \sqrt{1 + \sqrt{1 + \sqrt{1}}}, \sqrt{1 + \sqrt{1 + \sqrt{1 + \sqrt{1}}}}, \ldots$

Theory and Examples

99. The first term of a sequence is $x_1 = 1$. Each succeeding term is the sum of all those that come before it:

$$x_{n+1} = x_1 + x_2 + \cdots + x_n.$$

Write out enough early terms of the sequence to deduce a general formula for x_n that holds for $n \geq 2$.

100. A sequence of rational numbers is described as follows:

$$\frac{1}{1}, \frac{3}{2}, \frac{7}{5}, \frac{17}{12}, \ldots, \frac{a}{b}, \frac{a+2b}{a+b}, \ldots.$$

Here the numerators form one sequence, the denominators form a second sequence, and their ratios form a third sequence. Let x_n and y_n be, respectively, the numerator and the denominator of the nth fraction $r_n = x_n/y_n$.

a. Verify that $x_1^2 - 2y_1^2 = -1$, $x_2^2 - 2y_2^2 = +1$ and, more generally, that if $a^2 - 2b^2 = -1$ or $+1$, then

$$(a + 2b)^2 - 2(a + b)^2 = +1 \quad \text{or} \quad -1,$$

respectively.

b. The fractions $r_n = x_n/y_n$ approach a limit as n increases. What is that limit? (*Hint:* Use part (a) to show that $r_n^2 - 2 = \pm(1/y_n)^2$ and that y_n is not less than n.)

101. Newton's method The following sequences come from the recursion formula for Newton's method,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Do the sequences converge? If so, to what value? In each case, begin by identifying the function f that generates the sequence.

a. $x_0 = 1, \quad x_{n+1} = x_n - \dfrac{x_n^2 - 2}{2x_n} = \dfrac{x_n}{2} + \dfrac{1}{x_n}$

b. $x_0 = 1, \quad x_{n+1} = x_n - \dfrac{\tan x_n - 1}{\sec^2 x_n}$

c. $x_0 = 1, \quad x_{n+1} = x_n - 1$

102. a. Suppose that $f(x)$ is differentiable for all x in $[0, 1]$ and that $f(0) = 0$. Define sequence $\{a_n\}$ by the rule $a_n = nf(1/n)$. Show that $\lim_{n\to\infty} a_n = f'(0)$. Use the result in part (a) to find the limits of the following sequences $\{a_n\}$.

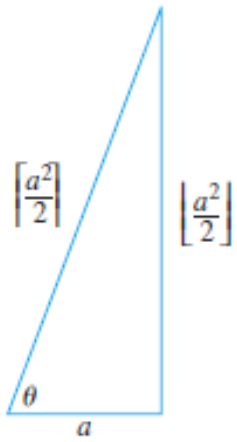
b. $a_n = n \tan^{-1} \dfrac{1}{n}$ **c.** $a_n = n(e^{1/n} - 1)$

d. $a_n = n \ln\left(1 + \dfrac{2}{n}\right)$

103. Pythagorean triples A triple of positive integers a, b, and c is called a **Pythagorean triple** if $a^2 + b^2 = c^2$. Let a be an odd positive integer and let

$$b = \left\lfloor \frac{a^2}{2} \right\rfloor \quad \text{and} \quad c = \left\lceil \frac{a^2}{2} \right\rceil$$

be, respectively, the integer floor and ceiling for $a^2/2$.



a. Show that $a^2 + b^2 = c^2$. (*Hint:* Let $a = 2n + 1$ and express b and c in terms of n.)

b. By direct calculation, or by appealing to the accompanying figure, find

$$\lim_{a\to\infty} \frac{\left\lfloor \frac{a^2}{2} \right\rfloor}{\left\lceil \frac{a^2}{2} \right\rceil}.$$

104. The *n*th root of *n*!

a. Show that $\lim_{n\to\infty}(2n\pi)^{1/(2n)} = 1$ and hence, using Stirling's approximation (Chapter 8, Additional Exercise 32a), that

$$\sqrt[n]{n!} \approx \frac{n}{e} \quad \text{for large values of } n.$$

T **b.** Test the approximation in part (a) for $n = 40, 50, 60, \ldots$, as far as your calculator will allow.

105. a. Assuming that $\lim_{n\to\infty}(1/n^c) = 0$ if c is any positive constant, show that

$$\lim_{n\to\infty} \frac{\ln n}{n^c} = 0$$

if c is any positive constant.

b. Prove that $\lim_{n\to\infty}(1/n^c) = 0$ if c is any positive constant. (*Hint:* If $\epsilon = 0.001$ and $c = 0.04$, how large should N be to ensure that $|1/n^c - 0| < \epsilon$ if $n > N$?)

106. The zipper theorem Prove the "zipper theorem" for sequences: If $\{a_n\}$ and $\{b_n\}$ both converge to L, then the sequence

$$a_1, b_1, a_2, b_2, \ldots, a_n, b_n, \ldots$$

converges to L.

107. Prove that $\lim_{n\to\infty}\sqrt[n]{n} = 1$.

108. Prove that $\lim_{n\to\infty} x^{1/n} = 1, (x > 0)$.

109. Prove Theorem 2.

110. Prove Theorem 3.

In Exercises 111–114, determine if the sequence is monotonic and if it is bounded.

111. $a_n = \dfrac{3n + 1}{n + 1}$

112. $a_n = \dfrac{(2n + 3)!}{(n + 1)!}$

113. $a_n = \dfrac{2^n 3^n}{n!}$

114. $a_n = 2 - \dfrac{2}{n} - \dfrac{1}{2^n}$

Which of the sequences in Exercises 115–124 converge, and which diverge? Give reasons for your answers.

115. $a_n = 1 - \dfrac{1}{n}$

116. $a_n = n - \dfrac{1}{n}$

117. $a_n = \dfrac{2^n - 1}{2^n}$ **118.** $a_n = \dfrac{2^n - 1}{3^n}$

119. $a_n = ((-1)^n + 1)\left(\dfrac{n+1}{n}\right)$

120. The first term of a sequence is $x_1 = \cos(1)$. The next terms are $x_2 = x_1$ or $\cos(2)$, whichever is larger; and $x_3 = x_2$ or $\cos(3)$, whichever is larger (farther to the right). In general,

$$x_{n+1} = \max\{x_n, \cos(n+1)\}.$$

121. $a_n = \dfrac{1 + \sqrt{2n}}{\sqrt{n}}$ **122.** $a_n = \dfrac{n+1}{n}$

123. $a_n = \dfrac{4^{n+1} + 3^n}{4^n}$

124. $a_1 = 1, \quad a_{n+1} = 2a_n - 3$

125. The sequence $\{n/(n+1)\}$ has a least upper bound of 1 Show that if M is a number less than 1, then the terms of $\{n/(n+1)\}$ eventually exceed M. That is, if $M < 1$ there is an integer N such that $n/(n+1) > M$ whenever $n > N$. Since $n/(n+1) < 1$ for every n, this proves that 1 is a least upper bound for $\{n/(n+1)\}$.

126. Uniqueness of least upper bounds Show that if M_1 and M_2 are least upper bounds for the sequence $\{a_n\}$, then $M_1 = M_2$. That is, a sequence cannot have two different least upper bounds.

127. Is it true that a sequence $\{a_n\}$ of positive numbers must converge if it is bounded from above? Give reasons for your answer.

128. Prove that if $\{a_n\}$ is a convergent sequence, then to every positive number ϵ there corresponds an integer N such that for all m and n,

$$m > N \quad \text{and} \quad n > N \quad \Rightarrow \quad |a_m - a_n| < \epsilon.$$

129. Uniqueness of limits Prove that limits of sequences are unique. That is, show that if L_1 and L_2 are numbers such that $a_n \to L_1$ and $a_n \to L_2$, then $L_1 = L_2$.

130. Limits and subsequences If the terms of one sequence appear in another sequence in their given order, we call the first sequence a **subsequence** of the second. Prove that if two subsequences of a sequence $\{a_n\}$ have different limits $L_1 \neq L_2$, then $\{a_n\}$ diverges.

131. For a sequence $\{a_n\}$ the terms of even index are denoted by a_{2k} and the terms of odd index by a_{2k+1}. Prove that if $a_{2k} \to L$ and $a_{2k+1} \to L$, then $a_n \to L$.

132. Prove that a sequence $\{a_n\}$ converges to 0 if and only if the sequence of absolute values $\{|a_n|\}$ converges to 0.

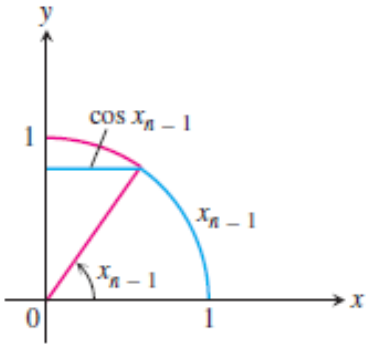
133. Sequences generated by Newton's method Newton's method, applied to a differentiable function $f(x)$, begins with a starting value x_0 and constructs from it a sequence of numbers $\{x_n\}$ that under favorable circumstances converges to a zero of f. The recursion formula for the sequence is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

a. Show that the recursion formula for $f(x) = x^2 - a$, $a > 0$, can be written as $x_{n+1} = (x_n + a/x_n)/2$.

T **b.** Starting with $x_0 = 1$ and $a = 3$, calculate successive terms of the sequence until the display begins to repeat. What number is being approximated? Explain.

T **134. A recursive definition of $\pi/2$** If you start with $x_1 = 1$ and define the subsequent terms of $\{x_n\}$ by the rule $x_n = x_{n-1} + \cos x_{n-1}$, you generate a sequence that converges rapidly to $\pi/2$. **(a)** Try it. **(b)** Use the accompanying figure to explain why the convergence is so rapid.



COMPUTER EXPLORATIONS

Use a CAS to perform the following steps for the sequences in Exercises 135–146.

a. Calculate and then plot the first 25 terms of the sequence. Does the sequence appear to be bounded from above or below? Does it appear to converge or diverge? If it does converge, what is the limit L?

b. If the sequence converges, find an integer N such that $|a_n - L| \le 0.01$ for $n \ge N$. How far in the sequence do you have to get for the terms to lie within 0.0001 of L?

135. $a_n = \sqrt[n]{n}$

136. $a_n = \left(1 + \dfrac{0.5}{n}\right)^n$

137. $a_1 = 1, \quad a_{n+1} = a_n + \dfrac{1}{5^n}$

138. $a_1 = 1, \quad a_{n+1} = a_n + (-2)^n$

139. $a_n = \sin n$

140. $a_n = n \sin \dfrac{1}{n}$

141. $a_n = \dfrac{\sin n}{n}$

142. $a_n = \dfrac{\ln n}{n}$

143. $a_n = (0.9999)^n$

144. $a_n = (123456)^{1/n}$

145. $a_n = \dfrac{8^n}{n!}$

146. $a_n = \dfrac{n^{41}}{19^n}$